

CHENYU GAO

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SUMMARY

Applied ML scientist and final-year ECE Ph.D. at Vanderbilt, graduating Nov 2026. Industry experience at TikTok (ByteDance) and insitro spanning LLM post-training, reward modeling for video generation models, multimodal representation learning, and generative models. Built large-scale ML systems, evaluation pipelines, and production-quality signals with measurable platform impact. Author of 40+ peer-reviewed publications, including 8 first/co-first-author works, and a pending U.S. patent application.

TECHNICAL EXPERTISE

Core ML: LLM Post-Training, SFT, RL/GRPO, Reward Modeling for Generative Video, Diffusion Models, Vision-Language Models, Multimodal Representation Learning, Self-Supervised Learning, Computer Vision

Data Infrastructure & Evaluation: Large-Scale Data Organization & Processing, Human/Model Evaluation & Benchmarking

Engineering: Python, PyTorch, Hugging Face, Slurm/HPC Workflows, Docker/Singularity, AWS EC2/S3, Git

EXPERIENCE

Machine Learning Scientist Intern

TikTok (ByteDance)

San Jose, CA

Jan 2026 – May 2026

- **Latent-space reward modeling for video generation:** Developed a DiT-native reward model targeting spatiotemporal consistency and multi-object interaction fidelity in generated videos.
- Proposed an early-step reward-evaluation strategy that scores VAE latents directly, bypassing VAE decoding and reducing required DiT steps by 75% by exploiting early stabilization of video layouts during denoising.
- Designed a *Semantic Adapter* to project VAE latents into a semantic feature space distilled from a vision foundation model, enabling object-centric evaluation of generated-video dynamics without full video decoding.
- Validated the lightweight reward model against large VLM judges, achieving stronger agreement with human ratings on target video qualities, and used it as the reward signal for GRPO-style RL post-training of video generation models.
- **LLM-driven video scoring:** Built an SFT pipeline to adapt a base LLM for multidimensional video-quality scoring, distilling rubric-guided rationales into concise outputs for higher-QPS serving.
- Integrated LLM-derived video-quality features into a production recommendation pipeline, driving a 4.15% increase in feature penetration and platform-wide lifts in publish frequency (+0.03%) and search engagement (+0.019%).

Data Science and Machine Learning Intern

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South San Francisco, CA

June 2025 – Aug 2025

- **Image-genetics self-supervised learning:** Designed a multimodal SSL framework for target discovery, combining masked image modeling with SNP-guided contrastive alignment from whole-genome sequencing data.
- **LLM-assisted genomic feature selection:** Built a human-in-the-loop LLM/ontology pipeline that reduced 1B+ genome-wide SNP candidates to a curated panel of hundreds, linking published genetic associations to biomedical ontology terms and prioritizing image-observable phenotypes for multimodal SSL.
- Converted learned image embeddings into disease-relevant quantitative phenotypes via regression and clustering, enabling GWAS in imaging cohorts to recover candidate genetic associations and surface target-discovery hypotheses.

Graduate Research Assistant

Vanderbilt University

Nashville, TN

July 2022 – Present

- **Large-scale ML infrastructure:** Built containerized processing, harmonization, and QC workflows for 100,000+ MRI scans from 40+ datasets across local and Slurm/HPC environments, enabling reproducible multi-cohort ML studies.
- **Generative AI privacy and safety:** Led MRI de-identification studies evaluating the privacy-utility tradeoff of defacing methods; built a cascaded diffusion-model red-team pipeline that reconstructs high-resolution 3D facial geometry from defaced head MRI, exposing re-identification risk and motivating privacy safeguards for medical data sharing.
- **Multimodal predictive modeling:** Developed brain-age models from diffusion and structural MRI across 13,398 participants and 12 datasets, leveraging micro- and macrostructural features with differential sensitivity across stages of neurodegenerative disease to support early detection; this work resulted in a U.S. patent application.
- **Generative modeling:** Co-developed conditional GAN models to extend incomplete MRI fields-of-view, rescuing otherwise unusable scans and increasing effective dataset size by 10%; also co-developed a diffusion model for joint modeling of imaging and clinical metadata.

Graduate Research Assistant

Johns Hopkins University

Baltimore, MD

Dec 2020 – May 2022

- **Continual learning:** Extended a lifelong learning algorithm from vision to speech and validated omnidirectional knowledge transfer on a spoken-digit benchmark, contributing to an open-source library and a publication in *IEEE TPAMI*.
- **Medical image analysis:** Benchmarked deep learning and computer vision methods for MRI analysis, establishing baselines for subsequent work on MRI privacy, defacing, and data utility evaluation.

EDUCATION

Vanderbilt University

Doctor of Philosophy in Electrical & Computer Engineering

Nashville, TN

July 2022 – Nov 2026 (expected)

Johns Hopkins University

Master of Science in Biomedical Engineering

Baltimore, MD

Aug 2020 – May 2022

Sun Yat-sen University

Bachelor of Science in Biomedical Engineering

Guangzhou, China

Aug 2016 – June 2020

SELECTED PUBLICATIONS & PATENT

Full list: 40+ peer-reviewed publications, including 8 first/co-first-author works; see Google Scholar.

- S1. **C Gao***, K Xu*, et al. “Pitfalls of defacing whole-head MRI: re-identification risk with diffusion models and compromised research potential.” *Computers in Biology and Medicine*. 2025. [**Diffusion-model privacy red-team; generative AI safety**]
- S2. **C Gao**, et al. “Brain age identification from diffusion MRI synergistically predicts neurodegenerative disease.” *Imaging Neuroscience*. 2025. [**Multimodal predictive modeling; patent-backed disease-risk prediction**]
- S3. **Patent: C Gao**, BA Landman, ME Kim. “System and Method of Brain Age Identification for Predicting Neuro-Degenerative Disease.” U.S. Patent Application No. 19/345,706. [**Lead inventor; applied predictive ML**]
- S4. **C Gao**, et al. “Field-of-view extension for brain diffusion MRI via deep generative models.” *Journal of Medical Imaging*. 2024. [**Conditional generative modeling**]
- S5. Y Liu, **C Gao**, et al. “MetaVoxel: Joint Diffusion Modeling of Imaging and Clinical Metadata.” *Medical Imaging with Deep Learning*. 2026. [**Multimodal diffusion modeling**]
- S6. JT Vogelstein, J Dey, ..., **C Gao**, et al. “Simple Lifelong Learning Machines.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2025. [**Continual learning**]
- S7. ME Kim, **C Gao**, et al. “White matter micro- and macrostructure brain charts for the human lifespan.” *Nature*. 2026. [**Population-scale ML infrastructure; normative brain-chart modeling**]